Practical Laboratory 2 Report

OPNsense Firewall Policy Testing, Logging and Packet Analysis

WADF105 – Network Security Fundamentals

Student Name: Abdul Fatawu Umar
Course: WADF105 Network Security Fundamentals
Laboratory: Practical Laboratory 2 – OPNsense Firewall Policy Testing, Logging and Packet Analysis
Date : 06th october 2026
VMs Used: icdfa-nslab-client-v1 and icdfa-nslab-firewall-v1

---

1. Introduction

In this laboratory, I configured and tested firewall policies using OPNsense. The objective was to understand how firewall rules are processed, how specific traffic can be blocked while allowing other traffic, and how firewall decisions can be verified using logs and packet captures.

The laboratory involved creating temporary firewall rules to block ICMP traffic to a specific destination and HTTP traffic on TCP port 80. The effects of these policies were validated using Ubuntu network tests, OPNsense firewall logs, and Wireshark packet captures. Finally, the environment was restored to its original state to ensure a clean configuration for subsequent exercises.

---

2. Learning Objectives

The objectives of this laboratory were to:

- Understand top-down firewall rule processing.
- Create and test specific firewall rules.
- Configure protocol, source, destination, and port filtering.
- Examine firewall logs generated by OPNsense.
- Correlate firewall logs with packet capture evidence.
- Restore the network to its original operational state.

---

3. Part A – Baseline Verification

Before any firewall policies were introduced, connectivity was verified to ensure the network was functioning correctly.

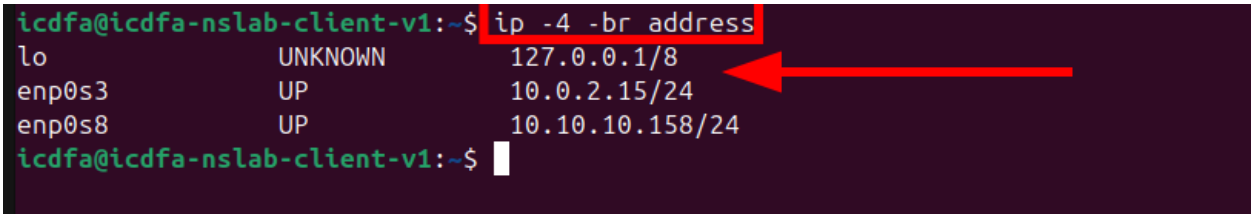
Verify IP Address

Command:

ip -4 -br address

Result:

The Ubuntu client received a valid IPv4 address from the 10.10.10.0/24 LAN network.



Verify Default Route

Command:

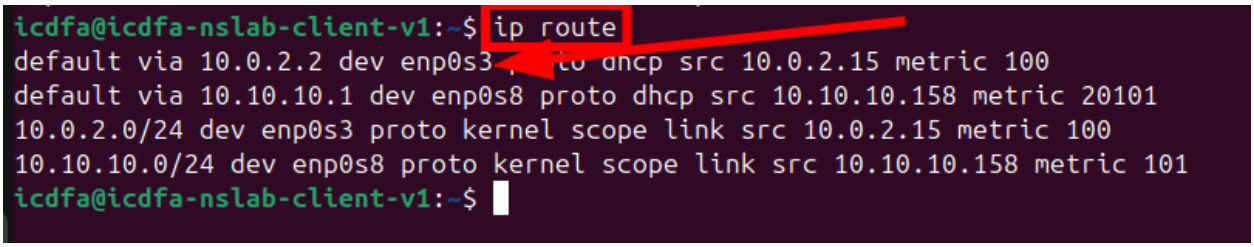
ip route

Result:

The default gateway was configured as:

10.10.10.1

which is the OPNsense LAN interface.



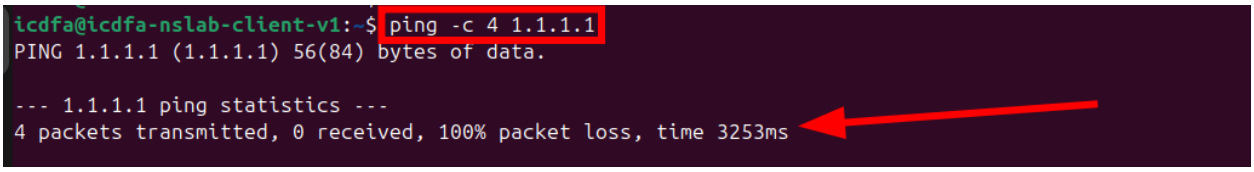
Test Internet Connectivity

Command:

ping -c 4 1.1.1.1

Result:

The Ubuntu client successfully reached the external host with 0% packet loss.



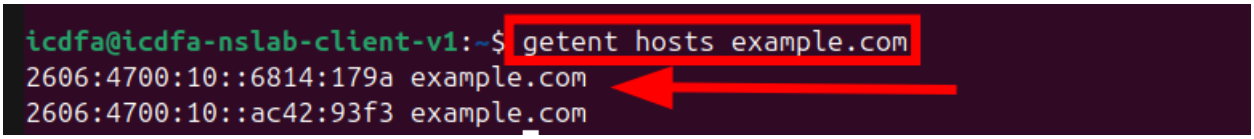
Test DNS Resolution

Command:

getent hosts example.com

Result:

The domain name was successfully resolved to an IP address.



Test HTTP and HTTPS

Commands:

curl -I http://example.com

curl -I https://example.com

``

Result:

Both HTTP and HTTPS returned valid responses, confirming the baseline network configuration was functioning normally.



4. Part B – Review Firewall Rule Order

The LAN rules section of OPNsense was examined.

Navigation:

Firewall → Rules → LAN

`

The existing rule:

Default allow LAN to any

was identified.

I observed that OPNsense processes firewall rules from top to bottom. Therefore, any block rule intended to override the default allow policy must be positioned above the allow rule.

---

5. Part C – Blocking ICMP Traffic

A new firewall rule was created to block ICMP traffic from the LAN network to host 1.1.1.1.

Configuration:

Field	Value
Action	Block
Interface	LAN
Direction	In

Field	Value
IP Version	IPv4
Protocol	ICMP
Source	LAN Network
Destination	1.1.1.1
Logging	Enabled
Description	LAB2 BLOCK ICMP TO 1.1.1.1

The rule was placed above the default allow rule and changes were applied.

ICMP Testing

Command:

ping -c 4 1.1.1.1

Result:

The ping requests failed and showed 100% packet loss.

Verify Other Services

Commands:

getent hosts example.com

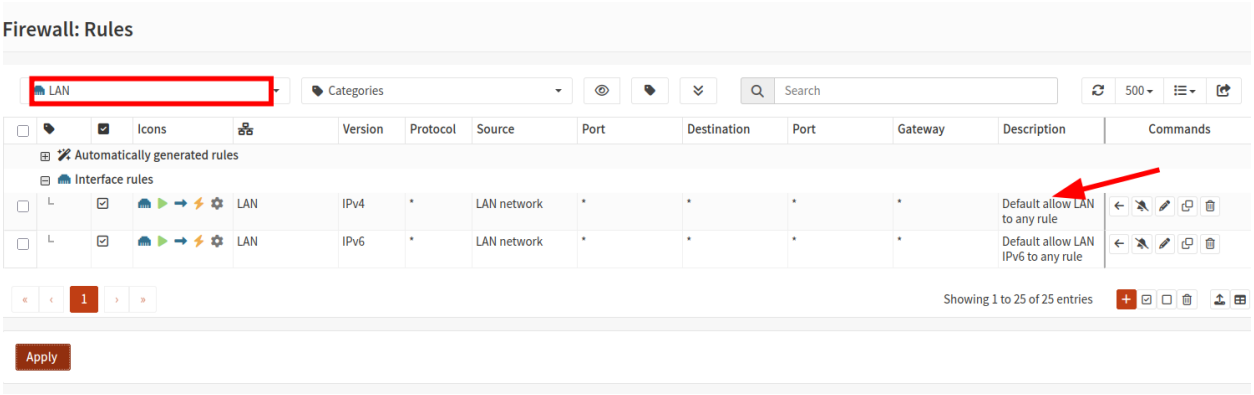
curl -I https://example.com

Result:

DNS and HTTPS continued to function successfully.

This demonstrated that only ICMP traffic was filtered while other protocols were still permitted.

```
icdfa@icdfa-nslab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 14:49:22 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 9367
cf-cache-status: HIT
cf-ray: a46578e96bb29334-ACC
alt-svc: h3=":443"; ma=86400
```



6. Part D – Blocking HTTP Traffic

A second firewall rule was created.

Configuration:

Field	Value
Action	Block
Interface	LAN
Direction	In
Protocol	TCP
Source	LAN Network

Field	Value
Destination	Any
Destination Port	80 (HTTP)
Logging	Enabled
Description	LAB2 BLOCK OUTBOUND HTTP

The rule was placed above the default allow rule.

HTTP Testing

Command:

curl --max-time 10 -I http://example.com

`

Result:

The request failed because TCP port 80 was blocked.

HTTPS Testing

Command:

curl --max-time 10 -I https://example.com

Result:

The HTTPS request completed successfully.

This confirmed that only unencrypted web traffic was blocked while encrypted HTTPS communication remained available.

7. Part E – Firewall Log Analysis

Navigation:

Firewall → Log Files → Live View

After generating ICMP and HTTP traffic, corresponding log entries appeared.

Information observed included:

- Timestamp
- Interface
- Source IP
- Destination IP
- Protocol
- Destination Port
- Action (Block)
- Matched Rule

Blocked ICMP traffic matched:

LAB2 BLOCK ICMP TO 1.1.1.1

Blocked HTTP traffic matched:

LAB2 BLOCK OUTBOUND HTTP

The traffic matched the specific block rules because they were evaluated before the default allow rule.

8. Part F – Wireshark Packet Analysis

Wireshark was launched on Ubuntu and a capture was started on the active network interface.

ICMP Analysis

Filter:

icmp && ip.addr == 1.1.1.1

Observation:

Echo Requests left the Ubuntu client but no Echo Replies returned.

This indicated the firewall blocked the traffic.

HTTP Analysis

Filter:

tcp.dstport == 80

Observation:

TCP SYN packets were observed along with retransmissions.

No successful connection was established.

HTTPS Analysis

Filter:

tcp.port == 443

Observation:

A complete TCP handshake occurred:

SYN

SYN-ACK

ACK

TLS traffic followed, indicating a successful encrypted connection.

9. Part G – State Table and NAT Inspection

Navigation:

Firewall → Diagnostics → States

A new HTTPS connection was generated and the state table was refreshed.

The table displayed:

- Client private address
- External destination
- Translated WAN address

This confirmed that the connection was tracked correctly by the firewall.

Outbound NAT Verification

Navigation:

Firewall → NAT → Outbound

Observation:

The firewall was operating in Automatic Outbound NAT mode.

This mechanism translated the private 10.10.10.0/24 addresses into a routable address on the WAN interface.

No configuration changes were made.

10. Part H – Restoring Normal Connectivity

The two laboratory rules were disabled:

LAB2 BLOCK ICMP TO 1.1.1.1

LAB2 BLOCK OUTBOUND HTTP

Changes were applied.

Connectivity Verification

Commands:

ping -c 4 1.1.1.1

curl -I http://example.com

curl -I https://example.com

Results:

All tests completed successfully.

This demonstrated that normal network functionality had been restored.

11. Analysis Questions

38. Why must the specific block rules be placed above the broad allow rule?

OPNsense processes firewall rules from top to bottom. If the allow rule appears before a block rule, matching traffic is permitted and the block rule is never reached.

39. Which five packet attributes are most useful when explaining a firewall decision?

1. Source IP address
2. Destination IP address
3. Protocol
4. Source port
5. Destination port

40. Why did blocking ICMP not block HTTPS?

ICMP and HTTPS use different protocols. The firewall rule specifically blocked ICMP traffic only, while HTTPS traffic using TCP port 443 continued to be allowed.

41. What difference did you observe between blocked TCP port 80 traffic and permitted TCP port 443 traffic?

Blocked TCP port 80 traffic showed repeated SYN retransmissions with no response. TCP port 443 traffic completed the TCP handshake and successfully exchanged encrypted TLS traffic.

42. What role does outbound NAT play when the Ubuntu client uses a private IPv4 address?

Outbound NAT translates the private 10.10.10.x address into the firewall's WAN address, allowing systems on the internet to communicate with the client.

43. Why is restoring the original state an important part of a controlled security laboratory?

Restoring the original configuration ensures future laboratories begin with a known-good baseline and prevents previous security policies from affecting later exercises.

12. Conclusion

This laboratory successfully demonstrated the implementation and testing of firewall policies in OPNsense. Specific ICMP and HTTP traffic was blocked while DNS and HTTPS traffic remained operational. Firewall logs and Wireshark packet captures provided evidence of how traffic was filtered. The results confirmed the importance of rule order, logging, and packet analysis in network security operations. Finally, all temporary rules were disabled and normal connectivity was restored, leaving the laboratory environment in a stable state.

Ethical assessment: This activity is **ethical and not immoral** because it involves defensive firewall administration, network monitoring, and security policy testing within an authorized training environment.